$f_{(X,Y)}(x,y) = \begin{cases} \frac{1}{5}(x+y+1), & x \in [0,1],\ y \in [0,2] \\ 0 \text{ rest} \end{cases}$ (24)

$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dy = \frac{1}{5}\int_0^2 (x+y+1)\,dy =$

$= \frac{1}{5}\left[x\cdot\int_0^2 dy + \int_0^2 y\,dy + \int_0^2 dy\right] =$

$= \frac{1}{5}\left[x\cdot y\Big|_0^2 + \frac{y^2}{2}\Big|_0^2 + y\Big|_0^2\right] =$

$= \frac{1}{5}\left[2x + 2 + 2\right] = \frac{2}{5}(x+2)$

$x \in [0,1]$

$f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dx =$

$= \frac{1}{5}\int_0^1 (x+y+1)\,dx = \frac{1}{5}\left[\int_0^1 x\,dx + y\int_0^1 dx + \int_0^1 dx\right]$

$= \frac{1}{5}\left[\frac{x^2}{2}\Big|_0^1 + y\cdot x\Big|_0^1 + x\Big|_0^1\right] =$

$= \frac{1}{5}\left[\frac{1}{2} + y + 1\right] = \frac{1}{5}\left(y + \frac{3}{2}\right)$

$y \in [0,2]$

X, Y indep dacă:

$f_{(X,Y)}(x,y) \overset{?}{=} f_X(x)\cdot f_Y(y)$

$\frac{1}{5}(x+y+1) \neq \frac{2}{5}(x+2)\cdot\frac{1}{5}\left(y+\frac{3}{2}\right)$

nu sunt indep.

$F_X(x)$, ştiu $f_X(x) = \frac{2}{5}(x+2)$

$x \in [0,1]$

$\int_{-\infty}^{x} f_X(t)\,dt$



$x \le 0$.

$$F_X(x) = \begin{cases} 0 & x \le 0 \\ \int_0^x \frac{2}{5}(t+2)\,dt & x \in [0,1] \\ 1 & x > 1 \end{cases}$$

$x \le 0$

$$F_X(x) = \begin{cases} 0 & x \le 0 \\ \frac{x^2}{5} + \frac{4}{5}x & x \in [0,1] \\ 1 & x \ge 1 \end{cases}$$

$$F_Y(y) = \int_{-\infty}^{y} f_Y(t)\,dt = \int_{-\infty}^{y} \frac{1}{5}(t+\tfrac{3}{2})\,dt$$

$x \le 0$, $y \le 0$

$x \in [0,1]$, $y \in [0,2]$

$$F_{(X,Y)}(x,y) = \begin{cases} 0 & \\ \int_0^x \int_0^y \frac{1}{5}(u+v+1)\,du\,dv & \end{cases}$$

$$\frac{1}{5}\left[\int_0^x \int_0^y (u+v+1)\,du\,dv\right] = \frac{1}{5}\left[\int_0^x \int_0^y u\,du\,dv + \int_0^x \int_0^y v\,du\,dv + \int_0^x \int_0^y du\,dv\right]$$

$$= \frac{1}{5}\left[\left(\int_0^x u\,du\right)\cdot\left(\int_0^y dv\right) + \left(\int_0^x du\right)\cdot\left(\int_0^y v\,dv\right) + \int_0^x du \int_0^y dv\right] =$$

$$= \frac{1}{5}\left[\frac{u^2}{2}\Big|_0^x \cdot v\Big|_0^y + u\Big|_0^x \cdot \frac{v^2}{2}\Big|_0^y + u\Big|_0^x \cdot v\Big|_0^y \right] =$$

$$= \frac{1}{5}\left[\frac{x^2}{2} \cdot y + x \cdot \frac{y^2}{2} + x \cdot y \right] =$$

$$= \frac{xy}{5}\left[\frac{x}{2} + \frac{y}{2} + 1 \right] = \frac{xy}{10}(x+y+2)$$

$x \le 0,\ y \le 0.$

$$F_{(x,y)}(x,y) \begin{cases} 0 & \\ \frac{xy}{10}(x+y+2), & x \in [0,1],\ y \in [0,2] \\ \frac{y}{10}(y+3), & x \in (1,\infty),\ y \in [0,2] \\ \frac{x}{5}(x+4), & x \in (0,1],\ y > 2 \\ 1 & \longrightarrow x > 1,\ y > 2. \end{cases}$$

$$M(X \mid Y=y) = \int_{-\infty}^{\infty} x\, f_{(x \mid Y=y)}(x \mid y)\, dx$$

$$f(x \mid y) = \frac{f(x,y)}{f_y(y)} = \frac{\frac{1}{5}(x+y+1)}{\frac{1}{5}(y+\frac{3}{2})} =$$

$$= \frac{x+y+1}{y+3/2}$$

$$M(X \mid Y=y) = \int_0^1 \frac{x(x+y+1)}{y+3/2}\, dx = \frac{1}{y+3/2} \int_0^1 x(x+y+1)\, dx$$

Proiect

- formula prob. clasică
- ——————— axiomatică ; Prob condiț.
- definiții
- scheme clasice
- FPT
- F Bayes
- Repartiții de prob. < discrete / continue
- val medii / disp.
- fc. repartiție
- v. aleatori bidimensionali < discret / cont.
- fc. generatoare, fc. caract.
- Enunț — ce înseamnă LNM } TLC }.

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